# Context-Aware Cross-Slice Transformer Framework for Automated Glioma Segmentation Using Multi-Modal MRI

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# BONAFIDE CERTIFICATE

This is to certify that the project report entitled "Context-Aware Cross-Slice Transformer Framework for Automated Glioma Segmentation Using Multi-Modal MRI", submitted by Harinandan G (AM.SC.U4CSE23122), Nived G Nair (AM.SC.U4CSE23139), Satvik Chaudhary (AM.SC.U4CSE23142) and Sunil Swaine (AM.SC.U4CSE23160), in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering at Amrita Vishwa Vidyapeetham, is a bonafide record of the work carried out by them under my guidance and supervision at Amrita School of Computing, Amritapuri, during Semester 7 of the academic year 2026 - 2027.

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# DECLARATION

We, Harinandan G (AM.SC.U4CSE23122), Nived G Nair (AM.SC.U4CSE23139), Satvik Chaudhary (AM.SC.U4CSE23142) and Sunil Swaine (AM.SC.U4CSE23160), hereby declare that the project entitled "Context-Aware Cross-Slice Transformer Framework for Automated Glioma Segmentation Using Multi-Modal MRI" is a record of our original work carried out under the guidance of Dr. Tintu Vijayan, Assistant Professor, Department of Computer Science and Engineering, Amrita Vishwa Vidyapeetham, Amritapuri. To the best of our knowledge, this work has not formed the basis for the award of any degree, diploma, fellowship or similar qualification to any candidate at any university. All sources and contributions of others have been appropriately acknowledged.

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# ABSTRACT

Gliomas are the most common malignant primary brain tumours, and precise delineation of their sub-regions on magnetic resonance imaging (MRI) underpins surgical planning, radiotherapy targeting and treatment monitoring. Manual slice-by-slice delineation is slow and varies between experts, while automated two-dimensional (2D) networks process every slice independently and can produce boundaries that are inconsistent through the volume; fully three-dimensional (3D) networks address this but demand large GPU memory. This project addresses these problems in the context of United Nations Sustainable Development Goal 3 (SDG 3), *Good Health and Well-being*: reliable, lightweight automated segmentation can reduce diagnostic delay and radiologist workload, which matters most where specialist capacity is scarce. We adapt the cross-slice transformer idea, originally proposed for lymph-node segmentation in CT, to multi-modal brain MRI. The proposed 2.5D model encodes five neighbouring slices (T1, T1ce, T2 and FLAIR) with a shared U-Net encoder, fuses them with self-attention across the slice axis at the bottleneck, and decodes the centre slice into whole-tumour (WT), tumour-core (TC) and enhancing-tumour (ET) masks. We built a reproducible pipeline with patient-level splits, volumetric BraTS-style evaluation, an Optuna hyper-parameter search, and execution on the institutional Astra A100 cluster. In five-fold cross-validation on all 369 BraTS 2020 training patients, against a 2D U-Net baseline trained identically, the cross-slice model achieved a mean 3D Dice of 0.844 versus 0.839 (WT 0.904 vs 0.898, TC 0.840 vs 0.837, ET 0.789 vs 0.783). Paired Wilcoxon tests over the 369 patients show that the Dice gains are small but statistically significant ($p < 0.001$). The larger improvements are in boundary accuracy and slice consistency: 95th-percentile Hausdorff distance fell by 1.8 mm (WT) and 5.1 mm (ET), and slices with false-positive tumour fell by 34%. These results support cross-slice attention as a compact way to improve volumetric consistency; clinical benefit has not been evaluated and remains an expected outcome.

Keywords: Glioma segmentation; Cross-slice transformer; 2.5D deep learning; Multi-modal MRI; BraTS 2020; SDG 3

# Table of Contents

**Acknowledgement** i

**Abstract** ii

**List of Figures** v

**List of Tables** vi

**1 Introduction** 1

1.1 Background and motivation . . . . . 1

1.2 Objectives . . . . . 2

1.3 Scope and constraints . . . . . 3

1.4 Project report organization . . . . . 3

**2 Problem Definition** 4

2.1 Problem statement . . . . . 4

2.2 Existing system and limitations . . . . . 4

**3 Literature Review** 6

3.1 Background of related work . . . . . 6

3.2 Review scope and selection . . . . . 7

3.3 Descriptive review of literature works . . . . . 8

3.4 Comparison of reviewed literature . . . . . 13

**4 Requirements** 16

4.1 Functional requirements . . . . . 16

4.2 Non-functional requirements . . . . . 17

4.3 Hardware requirements . . . . . 17

4.4 Software and data requirements . . . . . 18

**5 Proposed Methodology** **20**

5.1 Design (architecture) or conceptual framework . . . . . 20

5.2 Proposed approach and procedure . . . . . 20

**6 Results and Analysis** **28**

6.1 Evaluation setup . . . . . 28

6.2 Evaluation criteria . . . . . 29

6.3 Result . . . . . 29

6.4 Comparative Analysis . . . . . 35

6.5 Discussion and Interpretation . . . . . 36

6.6 Limitations . . . . . 37

**7 Conclusion** **38**

7.1 Conclusion . . . . . 38

7.2 Future work . . . . . 39

**References** **40**

**A Source Code and Reproducibility** **43**

A.1 Setup and execution . . . . . 43

A.2 Supporting material . . . . . 44

# List of Figures

5.1 Architecture of the proposed 2.5D Cross-Slice U-Net (shapes for $K = 5$, base width 48) . . . . . 21
5.2 End-to-end workflow of the study, repeated for each of the five folds . . . . 22

6.1 Mean 3D Dice per cross-validation fold for the baseline and the proposed model . . . . . 30
6.2 Distribution of per-patient differences between the proposed model and the baseline over 369 patients (boxes show the interquartile range; outliers omitted) 32
6.3 HD95 boundary error per region (mean and standard deviation over the five folds) . . . . . 32
6.4 Training loss and validation mean 3D Dice per epoch on fold 0 . . . . . 33
6.5 Optuna search: validation Dice of completed trials coloured by slice count $K$ (left) and hyper-parameter importance (right) . . . . . 34

# List of Tables


3.1 Comparison of reviewed literature . . . . . 13

4.1 Hardware used in the project . . . . . 17
4.2 Software stack . . . . . 18
4.3 Patient-level five-fold partition of the 369 BraTS 2020 training patients . . . 19

5.1 Preliminary one-factor-at-a-time design study (mid-review protocol: 16 training volumes, 10 epochs, per-slice validation Dice) . . . . . 25
5.2 Issues found in the mid-review prototype (v1) and changes in the final system (v2) . . . . . 27

6.1 Five-fold cross-validation results on all 369 patients (mean $\pm$ standard deviation of the five fold averages; $\uparrow$ higher is better, $\downarrow$ lower is better) . . . . . 30
6.2 Per-fold test results (73–74 patients per fold) . . . . . 30
6.3 Per-fold boundary error for TC and ET and false-positive slices per patient . 31
6.4 Paired per-patient comparison over all 369 patients (difference = $K = 5$ minus $K = 1$; two-sided Wilcoxon signed-rank test) . . . . . 32
6.5 Completed Optuna trials (validation mean 3D Dice after 8 short epochs, fold 0) . . . . . 33
6.6 Supplementary long-training experiment with early stopping (folds in which both runs completed; training stops 40 epochs after the best validation epoch) 34
6.7 Context against published BraTS 2020 results (different evaluation sets; for orientation only) . . . . . 36

A.1 Run directories behind the reported results . . . . . 44

# Chapter 1

# Introduction

## 1.1 Background and motivation

Gliomas are the most frequent malignant primary tumours of the brain. They arise from glial cells and range from slowly growing lower-grade lesions to aggressive glioblastoma, and their clinical management depends heavily on imaging. Multi-modal magnetic resonance imaging (MRI) is the standard tool: native T1-weighted (T1), contrast-enhanced T1-weighted (T1ce), T2-weighted (T2) and T2 fluid-attenuated inversion recovery (FLAIR) sequences each highlight different tissue properties, and together they reveal the oedema surrounding the tumour, the necrotic and non-enhancing core, and the actively enhancing rim [1, 2]. Delineating these sub-regions is required for surgical planning, for defining radiotherapy target volumes and for measuring the response to treatment over time.

In routine practice the delineation is still largely manual. A radiologist or radiation oncologist traces the tumour boundary slice by slice through a volume of roughly 150 axial slices, which is time-consuming and subject to inter-observer variability; the BraTS benchmark itself was created partly because expert annotations of the same scan disagree noticeably [1]. Deep learning has made automated segmentation practical. The U-Net encoder–decoder [3] and its volumetric extension, the 3D U-Net [4], are the basis of most current systems, and the self-configuring nnU-Net framework [5] shows that a carefully engineered U-Net remains among the strongest approaches. Transformer-based models such as TransBTS [6] and Swin UNETR [7] add long-range context to these networks.

There is, however, a practical tension between context and cost. Purely two-dimensional (2D) networks are light and fast but treat every slice independently, although the true tumour changes smoothly from one slice to the next; their predictions can therefore jump between adjacent slices, producing fragmented sub-regions and inaccurate 3D boundaries. Fully three-dimensional (3D) networks capture inter-slice context directly but require large GPU memory and long training times, which limits their use where computing resources are modest. The 2.5D family sits between the two: it feeds a small stack of neighbouring

slices to a 2D-style network so that some volumetric context is available at a fraction of the cost. A recent example is the Cross-Slice Transformer (CST) of Abdikenov *et al.* [8], which learns how much each of five consecutive CT slices should contribute to the segmentation of the centre slice, and which achieved the best Dice among 2.5D models for lymph-node segmentation with fewer than 8 million parameters. That work was designed for single-channel CT and small lymph nodes; whether cross-slice attention helps multi-modal brain MRI and glioma sub-regions had not been tested. This project performs that adaptation and evaluates it rigorously.

**Relevance to the Sustainable Development Goals.** The project is aligned with United Nations Sustainable Development Goal 3 (SDG 3), *Good Health and Well-being* [9]. It relates most directly to Target 3.4, which aims to reduce premature mortality from non-communicable diseases (including cancer) through prevention and treatment, and to Target 3.8, which calls for access to quality essential health-care services for all. Brain-tumour treatment quality depends on accurate and timely delineation, yet expert neuroradiology time is limited, especially in resource-constrained hospitals. The problem addressed is therefore the cost and inconsistency of glioma delineation; the project's contribution is a compact segmentation model and an evaluation showing that cross-slice context improves boundary accuracy and slice-to-slice consistency without the memory demands of a full 3D network. The intended beneficiaries are radiologists and radiation oncologists, who could review and correct an automatic draft instead of tracing from scratch, and ultimately patients, through faster and more consistent treatment planning. These clinical benefits are *expected* outcomes: the work demonstrates technical performance on a public research dataset and has not been evaluated in a clinical workflow.

## 1.2 Objectives

The project has six objectives, each tied to an evaluation method and a deliverable. The first objective is to design and implement a 2.5D Context-Aware Cross-Slice Transformer that fuses neighbouring-slice information for multi-modal MRI glioma segmentation; the deliverable is a working PyTorch implementation whose cross-slice module can be switched off by setting the number of slices $K = 1$, which yields an otherwise identical 2D U-Net baseline. The second objective is to build a reproducible experimental pipeline with patient-level data splits and evaluation that follows the BraTS convention of volumetric, per-patient metrics; it is evaluated by verifying that no patient appears in more than one of the training, validation and test sets and that test patients are used only once, and its deliverable is the public code repository. The third objective is to measure the contribution of cross-slice context through a controlled comparison of $K = 5$ against $K = 1$ under identical training, using five-fold cross-validation over all 369 annotated BraTS 2020 patients, with the 3D Dice coefficient as the primary metric and paired Wilcoxon signed-rank tests to establish

significance. The fourth objective is to justify the training hyper-parameters systematically rather than by trial and error, through a one-factor-at-a-time design study and an automated Optuna search [10]. The fifth objective is to quantify 3D boundary quality and slice-to-slice consistency, the property that cross-slice attention is intended to improve, using the 95th-percentile Hausdorff distance (HD95) and a count of slices containing false-positive tumour. The sixth objective is to make the pipeline run on the institutional Astra GPU cluster, so that the full-scale experiments that were impossible on a 4 GB laptop GPU could be completed; the deliverable is a set of scheduler-ready job scripts and the resulting five-fold results.

## 1.3 Scope and constraints

The intended users of the system are researchers and, in a future clinical setting, radiologists who would review its output. The scope covers automated segmentation of the three standard BraTS tumour regions from pre-operative multi-modal MRI, the comparison of the proposed model with a matched baseline, and the associated hyper-parameter study. Tumour grading, molecular-marker prediction, survival prediction and any clinical deployment are excluded. The work uses only the publicly available BraTS 2020 training set; the official BraTS validation and test sets do not release ground-truth masks to participants outside the challenge, so performance is estimated by internal cross-validation rather than on the official leaderboard. The approach is deliberately 2.5D: a full 3D model was outside the time budget and would also remove the very inter-slice modelling that the project sets out to study.

The project operated under significant computational constraints. Development and the preliminary design study were carried out on a laptop with an NVIDIA RTX 3050 GPU (4 GB), which forced small batches and a restricted protocol. The final experiments were run on the Astra cluster during a time-limited access window of two lab days plus one night, with at most three concurrent GPU jobs because the facility is shared. Each configuration was therefore trained once per fold with a fixed seed; variation across random seeds is not measured. Finally, the system is a research prototype and has not been validated as a medical device.

## 1.4 Project report organization

Chapter 2 defines the problem and the limitations of existing systems. Chapter 3 reviews twelve relevant research works and compares them in a table. Chapter 4 lists the functional, non-functional, hardware, software and data requirements. Chapter 5 presents the proposed architecture, the training and inference procedure, the hyper-parameter justification and the evolution of the system from the mid-review prototype. Chapter 6 reports the evaluation setup, criteria, results and their analysis, including limitations. Chapter 7 concludes the report and outlines future work. The appendix describes the source code and how to reproduce the experiments.

# Chapter 2

# Problem Definition

## 2.1 Problem statement

Given a pre-operative multi-modal brain MRI examination consisting of co-registered T1, T1ce, T2 and FLAIR volumes, the task is to label every voxel as belonging to one of three nested tumour regions defined by the BraTS challenge: the *whole tumour* (WT), comprising oedema, necrotic and non-enhancing core and enhancing tumour; the *tumour core* (TC), comprising necrotic and non-enhancing core and enhancing tumour; and the *enhancing tumour* (ET). The input to the proposed model is a stack of $K$ consecutive axial slices, each with four channels, and the output is the three-region mask of the centre slice; applying the model to every slice produces a full 3D segmentation.

The specific research question is whether attention across neighbouring slices, adapted from CT lymph-node segmentation, makes 2.5D glioma segmentation more accurate and more consistent through the volume than an otherwise identical 2D network, and whether any improvement is statistically reliable across patients. The stakeholders are clinicians who need accurate and consistent tumour volumes, patients whose treatment depends on them, and researchers who need lightweight models that can be trained without large GPU clusters. Inconsistent slice-wise segmentations have practical consequences: fragmented or jagged tumour volumes distort volume measurements, inflate boundary errors that matter for radiotherapy margins, and force clinicians to spend time correcting automatic output.

## 2.2 Existing system and limitations

The current clinical system is manual or semi-automatic contouring by experts. It is accurate when time allows, but it is slow, depends on scarce specialist time and shows inter-rater variability [1]. Among automated systems, three families are relevant, each with an evidence-based limitation.

Two-dimensional networks such as the U-Net [3] and its attention-augmented variants seg-

ment each slice independently. They are efficient, but they ignore the anatomical continuity between neighbouring slices. In our own mid-review prototype, a 2D baseline produced on average 17.7 slices per patient that contained predicted tumour although the ground truth had none, and its boundary error (HD95) for the whole tumour was 17.9 mm, almost twice that of a cross-slice variant evaluated under the same conditions (Section 5.2.8).

Three-dimensional networks such as the 3D U-Net [4], nnU-Net [5], TransBTS [6] and Swin UNETR [7] model volumetric context directly and achieve the strongest results on BraTS, but they require large GPU memory and long training schedules. The base CST paper confirms this trade-off in a different domain: a 3D nnU-Net achieved the highest overall Dice, while the 2.5D CST matched much of its performance with a far smaller footprint [8]. Recent glioma models listed in Chapter 3 add multi-scale aggregation, CNN–transformer hybrids or multi-task heads [11, 12, 13], but none of them explicitly models how one slice relates to its neighbours.

Finally, the cross-slice mechanism that does model inter-slice relationships [8] was designed for single-channel CT and very small targets, and its fusion operates on intensity statistics of raw slices. Its value for four-channel MRI and for large, heterogeneous gliomas with nested sub-regions is unknown. The gap addressed by this project is therefore the adaptation of cross-slice attention to multi-modal glioma MRI, together with a controlled, statistically tested measurement of what it contributes in accuracy, boundary quality and slice consistency.

# Chapter 3

# Literature Review

## 3.1 Background of related work

Automated brain-tumour segmentation has developed over roughly a decade, largely around the Multimodal Brain Tumor Segmentation (BraTS) benchmark. The benchmark was introduced in 2012 and described by Menze *et al.* [1], and it was later extended with expert labels for public glioma collections [2] and with a long-running annual challenge [14]. Early BraTS editions were dominated by methods based on hand-crafted features and classical classifiers. Menze *et al.* reported that different algorithms performed best on different tumour subregions, that no single algorithm ranked at the top for all of them, and that fusing several good algorithms by hierarchical majority voting outperformed every individual algorithm [1]. The benchmark also standardised the three nested evaluation regions (whole tumour, tumour core and enhancing tumour) and the use of Dice and Hausdorff-based metrics, which this project follows.

The first generation of deep-learning approaches applied convolutional neural networks (CNNs) to two-dimensional patches. Pereira *et al.* [15] showed that a deep CNN with small $3 \times 3$ kernels, combined with intensity normalisation and data augmentation, segments gliomas effectively from multi-modal MRI. Havaei *et al.* [16] proposed a two-pathway architecture that captures local detail and wider context at the same time, and a cascaded variant that feeds the output of one network into a second. These networks classified the centre pixel of each patch and were therefore slow at inference. Fully convolutional encoder–decoder networks, in particular the U-Net [3], removed this limitation by segmenting a whole image in one forward pass. The Attention U-Net [17] later added attention gates that suppress irrelevant regions in the skip connections.

Three-dimensional networks then became the leading approach on BraTS. Kamnitsas *et al.* [18] introduced DeepMedic, an 11-layer dual-pathway 3D CNN with an efficient dense-training scheme and a fully connected conditional random field for post-processing, which processes adjacent patches of a 3D scan efficiently. The 3D U-Net [4] extended the encoder–

decoder design to volumes. Myronenko [19] won the BraTS 2018 challenge with a 3D encoder–decoder regularised by a variational auto-encoder branch, and Isensee *et al.* won BraTS 2020 by adapting the self-configuring nnU-Net [5] to the challenge with a small number of task-specific modifications [20]. These works established that volumetric context is valuable, but they also show the cost: 3D models process large patches, need substantial GPU memory and train for many epochs.

The most recent generation adds transformer-based global context. TransUNet [21] placed a vision transformer on top of a CNN encoder for 2D medical images, TransBTS [6] brought the same idea to 3D brain-tumour MRI, and UNETR [22] and Swin UNETR [7] used transformers as the main 3D encoder. In parallel, 2.5D methods have tried to obtain some volumetric context at 2D cost by giving a network several neighbouring slices. Roth *et al.* [23] showed early on that 2.5D views improve CNN-based lymph-node detection in CT. More recently, Niu *et al.* [24] found that a multi-channel 2.5D model outperforms a 2D model for glioma classification, and Abdikenov *et al.* [8] introduced attention across neighbouring slices for 2.5D CT segmentation. Most 2.5D methods simply stack neighbouring slices as input channels and leave it to the first convolution to mix them. Learned attention across slices, which decides how much each neighbour should contribute, has not been studied for multi-modal glioma MRI. This is the gap the reviewed works lead to, and the one addressed by this project.

The remainder of this chapter first states the scope and selection criteria of the detailed review, then discusses twelve selected works individually, and finally compares them in a table.

## 3.2 Review scope and selection

The review covers twelve primary research works. The search used Google Scholar, PubMed, Europe PMC, IEEE Xplore and the Nature and Frontiers portals with combinations of the keywords *glioma segmentation*, *brain tumour segmentation*, *BraTS*, *multi-modal MRI*, *transformer*, *2.5D*, *cross-slice* and *inter-slice context*. Works were selected if they were peer-reviewed, addressed glioma analysis on multi-modal MRI or introduced a segmentation architecture used as a reference in this field, and reported a quantitative evaluation. The selection spans 2015 to 2026 and has two parts. The first part consists of seven recent works (2024–2026): five on glioma analysis, which define the current state of the application area, our previous-semester reference paper, and the cross-slice transformer that is the base paper of this project. The second part consists of five foundational architectures (2015–2022) against which glioma segmentation methods are routinely compared. Bibliographic details were verified against the publishers' records; quantitative findings are taken from the papers' abstracts or full texts, and details that we could not verify are marked "Not reported".

## 3.3 Descriptive review of literature works

### 3.3.1 MM-MSCA-AF: multi-scale contextual aggregation and attention fusion

Aslam *et al.* (2025), in "Enhanced brain tumor segmentation in medical imaging using multi-modal multi-scale contextual aggregation and attention fusion" [11], investigate brain-tumour segmentation from multi-modal MRI, where tumour heterogeneity and variability across sequences make segmentation difficult. The work proposes the MM-MSCA-AF framework, which takes T1, T2, FLAIR and T1ce images, uses multi-scale contextual aggregation to obtain both global and fine-grained spatial features, and applies gated attention fusion to keep informative features and suppress noise. The study is set on the BraTS 2020 dataset. The authors evaluate the model with the Dice coefficient and report a Dice of 0.8158 for the necrotic tumour region and 0.8589 overall, outperforming U-Net, nnU-Net and Attention U-Net in their comparison. Limitations stated beyond the single-dataset evaluation are not reported in the abstract, and boundary metrics such as HD95 are not reported there. The model aggregates context within a slice or patch and does not explicitly model the relationship between neighbouring slices; this gap, together with its use of the same dataset as our project, makes it a direct point of comparison.

### 3.3.2 DeepGlioSeg: CNN–transformer parallel combination

Li *et al.* (2025), in "DeepGlioSeg: advanced glioma MRI data segmentation with integrated local-global representation architecture" [12], address the heterogeneity of gliomas and the imbalance between tumour regions, which hinder existing segmentation methods. They propose a U-shaped network whose core is a CNN–Transformer Parallel Combination (CTPC) module: parallel convolutional and transformer branches fuse local and global features. A region-based probability, computed from the ratio of tumour to background pixels, gives more weight to under-represented regions, and test-time augmentation with volume-constrained post-processing refines the output. The study uses three public glioma MRI datasets and one private clinical dataset. The authors report, from quantitative and qualitative comparisons, that DeepGlioSeg outperforms other state-of-the-art methods; the per-region values were not extracted for this review and are therefore not reported here. The authors do not state limitations in the abstract. The work shows the value of combining local and global representations and of post-processing, but it models context within the image rather than across slices, which is the dimension our project targets.

### 3.3.3 Deep ensemble learning for joint segmentation and grading

Wen *et al.* (2025), in "A deep ensemble learning framework for glioma segmentation and grading prediction" [13], observe that most methods treat glioma segmentation and risk-grade prediction as separate single tasks and that grading data are scarce. They propose a deep-ensemble framework built on U-Net that performs both tasks from pre-operative multi-modal MRI. The encoder uses asymmetric convolution and dual-domain attention to integrate information from the modalities, and a dual-branch decoder combines spatial features with global semantic information. A weighted composite adaptive loss balances the two objectives. The study is set on the BraTS dataset. The authors report that the method outperforms state-of-the-art approaches in both segmentation accuracy and grade prediction; exact values are not reported in the abstract. The architecture is convolutional with attention inside each image and does not model long-range dependencies between slices. It informs our project by showing that attention-augmented U-Nets remain a strong design for BraTS, while leaving inter-slice modelling open.

### 3.3.4 SSL-MISS-Net: learning from incomplete sequences and annotations

Song *et al.* (2025), in "An integrated MRI-based diagnostic framework for glioma with incomplete imaging sequences and imperfect annotations" [25], address multi-centre glioma diagnosis, where differing imaging protocols and testing standards lead to missing MRI sequences and missing labels. They propose SSL-MISS-Net, which handles missing input sequences through cross-modal self-supervised learning and missing labels through a missing-label synergistic strategy. The study uses multi-centre cohorts from six in-house datasets and three public repositories (2,238 patients), evaluated with five-fold cross-validation and two independent test sets. Compared with the next-best method, the framework improved integrated-diagnosis accuracy by 4% on validation and 10% on test data and increased the amount of clinically usable data by 256%. The task is diagnosis and molecular classification rather than voxel-level segmentation, so segmentation boundary quality is not applicable. The work is relevant because it shows the value of robustness to imperfect, real-world clinical data, a requirement our model would need to meet before clinical use.

### 3.3.5 CrossFormer 2.5D model with radiomics for IDH/TERT prediction

Niu *et al.* (2025), in "MRI transformer deep learning and radiomics for predicting IDH wild type TERT promoter mutant gliomas" [24], aim to identify IDH-wildtype, TERT-promoter-mutant gliomas from MRI and to stratify patient risk. They develop a multi-channel 2.5D deep-learning model and a 2D model, both based on the cross-scale attention vision trans-

former (CrossFormer), together with a radiomics model, and combine them in a stacking ensemble. Pre-operative T1CE and FLAIR images from 1,185 patients are analysed. The 2.5D model outperformed both the 2D and the radiomics models, with areas under the ROC curve (AUC) of 0.806–0.870, and the stacking model achieved the highest AUC of 0.855–0.904 across validation sets; Kaplan–Meier analysis showed significant survival differences between the predicted risk groups. The task is classification, so segmentation metrics are not applicable, and limitations are not reported in the abstract. Its finding that a 2.5D multi-slice input outperforms a 2D input on glioma MRI directly motivates our use of neighbouring-slice context for segmentation.

### 3.3.6 Multi-task deep learning pipeline for glioma molecular status, grade and prognosis

Wu *et al.* (2024), in "Biologically interpretable multi-task deep learning pipeline predicts molecular alterations, grade, and prognosis in glioma patients" [26], develop an end-to-end multi-task deep-learning pipeline that predicts IDH mutation, 1p/19q co-deletion and histological grade as auxiliary tasks and prognosis as the primary task, and they relate the predicted deep prognosis score to biological mechanisms. The study uses MRI from 2,776 patients: two private cohorts ($n = 1931$) for training and internal validation, and three public cohorts (TCGA, UCSF and EGD) for external validation. In external validation the pipeline reached AUCs of 0.892–0.903 for IDH status, 0.710–0.894 for 1p/19q status and 0.850–0.879 for grade, and C-indices of 0.723 (TCGA) and 0.671 (UCSF) for overall survival. This was the reference paper of our previous semester. Its pipeline depends on accurate tumour localisation, and the wide range of 1p/19q performance across cohorts illustrates how sensitive downstream analysis is to data quality. The work motivated our shift to the segmentation stage itself, since every downstream prediction depends on a reliable tumour mask.

### 3.3.7 Cross-Slice Transformer with adaptive slice fusion (base paper)

Abdikenov *et al.* (2026), in "Cross-slice transformer with adaptive slice fusion for lymph node segmentation in CT imaging" [8], address lymph-node segmentation in CT, which is difficult because nodes are small, contrast is low and foreground voxels make up below 1% of the volume. They propose the Cross-Slice Transformer (CST), a lightweight 2.5D architecture that takes five consecutive slices, extracts a statistical descriptor (mean, standard deviation, inter-slice difference and their interaction) from each slice, and applies transformer attention to these descriptors to learn adaptive slice-importance weights. The fused representation is decoded by a U-Net. The study uses two public TCIA datasets (296 and 206 patients) and compares CST with 2.5D U-Net, TransUNet and SwinUNet baselines and with a 3D nnU-Net. On the mediastinal dataset, CST achieved the highest Dice among 2.5D models ($0.543 \pm 0.044$) and the highest 2.5D recall (0.618) with only 7.97 M parameters,

against 25–99 M for the 2.5D baselines, while nnU-Net-3D achieved the highest overall Dice (0.589). Ablations attribute gains of $+0.006$, $+0.008$ and $+0.008$ Dice to adaptive attention, the interaction descriptor and per-case intensity normalisation respectively. The limitations are its single-channel CT setting and its lower Dice than the 3D model. It is the base of our project: we adapt its central idea, attention across neighbouring slices, to four-channel MRI and to large nested glioma sub-regions, applying the attention to learned bottleneck features rather than to intensity statistics.

### 3.3.8 U-Net

Ronneberger *et al.* (2015), in "U-Net: convolutional networks for biomedical image segmentation" [3], address the need for accurate segmentation from few annotated biomedical images. They propose the U-Net, a contracting path that captures context and a symmetric expanding path whose skip connections recover precise localisation, trained end to end with heavy elastic-deformation augmentation. The study uses electron-microscopy and light-microscopy cell-segmentation challenges from the ISBI workshops. The network won the ISBI 2015 cell-tracking challenge, with an intersection-over-union of 0.9203 on the PhC-U373 dataset and 0.7756 on DIC-HeLa, and it segments a $512 \times 512$ image in under a second on a GPU. The original work is 2D and does not address volumetric consistency. The U-Net is the backbone of both our proposed model and our baseline.

### 3.3.9 3D U-Net

Çiçek *et al.* (2016), in "3D U-Net: learning dense volumetric segmentation from sparse annotation" [4], extend the U-Net to volumetric data by replacing 2D operations with 3D convolutions, pooling and up-convolutions, and they show that dense 3D segmentation can be learned from a few annotated 2D slices. The study uses confocal microscopy volumes of the *Xenopus* kidney and covers both a semi-automatic setting (densifying sparse annotations) and a fully automatic setting (segmenting unseen volumes); the authors report intersection-over-union and show that the 3D network outperforms slice-wise 2D processing. The main limitation for clinical MRI is the memory cost of 3D feature maps, which constrains input size and batch size. The work establishes the benefit of inter-slice context, which our 2.5D model seeks at lower cost.

### 3.3.10 nnU-Net

Isensee *et al.* (2021), in "nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation" [5], argue that designing segmentation solutions is highly dependent on dataset properties and hardware. They propose nnU-Net, which automatically configures preprocessing, network architecture, training and post-processing for a new task from a set of fixed parameters, interdependent rules and empirical decisions. Without man-

ual intervention, nnU-Net surpassed most existing approaches, including highly specialised solutions, on 23 public datasets used in international segmentation competitions. Its limitation in our setting is computational: its full-resolution 3D configuration and long training schedule exceed the resources available to us for most of the project. It informs our work as the strongest standard reference and as evidence that careful training protocol and evaluation matter as much as architecture.

### 3.3.11 TransBTS

Wang *et al.* (2021), in "TransBTS: multimodal brain tumor segmentation using transformer" [6], aim to add long-range dependency modelling to 3D CNNs for brain-tumour segmentation. They encode the multi-modal MRI volume with a 3D CNN, model global context at the bottleneck with a transformer applied to the resulting tokens, and decode with a 3D CNN. The study uses the BraTS 2019 and BraTS 2020 datasets, evaluated with Dice and Hausdorff distance on the official validation sets. On the BraTS 2020 validation set the authors report Dice scores of 0.7873 (ET), 0.9009 (WT) and 0.8173 (TC). The method relies on 3D patches and their memory cost. Its idea of placing a transformer at the U-Net bottleneck is shared by our design, but our transformer attends across slices rather than across spatial positions of a 3D volume.

### 3.3.12 Swin UNETR

Hatamizadeh *et al.* (2022), in "Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images" [7], extend their earlier UNETR [22] by using a Swin transformer encoder with shifted-window self-attention to extract features at five resolutions from multi-modal 3D MRI, connected to a convolutional decoder through skip connections. The study is set in the BraTS 2021 challenge, evaluated with Dice and HD95, and the method was reported among the top-ranking approaches in the challenge validation phase. Like other 3D transformers it requires substantial GPU memory and training time. It represents the state of transformer-based 3D brain-tumour segmentation and indicates the accuracy that full volumetric context can reach when resources allow.

## 3.4 Comparison of reviewed literature

Table 3.1 compares the twelve reviewed works. Study context covers the data and setting, and missing details are marked "Not reported".

Table 3.1: Comparison of reviewed literature

| **Work and citation** | **Problem or objective** | **Approach and contribution** | **Study context** | **Evaluation criteria** | **Key findings** | **Limitations and gap** |
|---|---|---|---|---|---|---|
| Work 01 [11] Aslam, 2025 | Brain-tumour segmentation from multi-modal MRI | Multi-scale contextual aggregation and gated attention fusion (MM-MSCA-AF) | BraTS 2020; T1, T2, FLAIR, T1ce | Dice | Dice 0.8589 overall, 0.8158 necrotic region; better than U-Net, nnU-Net, Attention U-Net | No inter-slice modelling; HD95 not reported |
| Work 02 [12] Li, 2025 | Glioma segmentation under heterogeneity and region imbalance | CNN–transformer parallel module, region-probability weighting, TTA and volume-constrained post-processing | Three public datasets and one private clinical dataset | Quantitative and qualitative segmentation metrics | Reported superior to state-of-the-art methods; values not reported here | Context within images, not across slices |
| Work 03 [13] Wen, 2025 | Joint glioma segmentation and risk grading | U-Net ensemble with asymmetric convolution, dual-domain attention, dual-branch decoder, adaptive composite loss | BraTS | Segmentation accuracy; grading accuracy | Reported better than state-of-the-art in both tasks; values not reported in abstract | Convolutional; no long-range inter-slice dependency |

| Work and citation | Problem or objective | Approach and contribution | Study context | Evaluation criteria | Key findings | Limitations and gap |
|---|---|---|---|---|---|---|
| Work 04 [25] Song, 2025 | Diagnosis with missing sequences and labels | SSL-MISS-Net: cross-modal self-supervision and missing-label strategy | 2,238 patients, nine multi-centre cohorts; five-fold CV and two test sets | Diagnostic accuracy | +4% (validation) and +10% (test) accuracy over next-best; +256% usable data | Classification, not voxel-level segmentation |
| Work 05 [24] Niu, 2025 | Predict IDH-wt, TERTp-mutant gliomas; risk stratification | CrossFormer 2.5D and 2D models plus radiomics in a stacking ensemble | 1,185 patients; T1CE and FLAIR | AUC; Kaplan–Meier, log-rank | 2.5D AUC 0.806–0.870 beat 2D and radiomics; stacking AUC 0.855–0.904 | Classification only; supports value of 2.5D input |
| Work 06 [26] Wu, 2024 | Predict molecular status, grade and prognosis | End-to-end multi-task deep learning with interpretable prognosis score | 2,776 patients; private training, three public external cohorts | AUC; C-index | External AUC 0.892–0.903 (IDH), 0.710–0.894 (1p/19q), 0.850–0.879 (grade); C-index 0.723 / 0.671 | Depends on tumour localisation; segmentation not the focus |
| Work 07 [8] Abdikenov, 2026 | Lymph-node segmentation in CT | 2.5D Cross-Slice Transformer: per-slice descriptors with transformer attention for adaptive slice fusion | Two TCIA CT datasets (296 and 206 patients) | Dice, recall, node detection rate, parameters | Best 2.5D Dice $0.543 \pm 0.044$ with 7.97 M parameters; nnU-Net-3D 0.589 | Single-channel CT, small targets; untested on multi-modal MRI |

| Work and citation | Problem or objective | Approach and contribution | Study context | Evaluation criteria | Key findings | Limitations and gap |
|---|---|---|---|---|---|---|
| Work 08 [3] Ronneberger, 2015 | Segmentation from few annotations | U-Net encoder–decoder with skip connections and elastic augmentation | ISBI EM and cell-tracking challenges | IoU; warping error | Won ISBI 2015 cell tracking: IoU 0.9203 (PhC-U373), 0.7756 (DIC-HeLa) | 2D only; no volumetric consistency |
| Work 09 [4] Çiçek, 2016 | Volumetric segmentation from sparse labels | 3D U-Net with 3D convolutions; learns from sparsely annotated slices | *Xenopus* kidney confocal volumes | IoU | 3D network outperformed slice-wise 2D processing | High memory cost of 3D feature maps |
| Work 10 [5] Isensee, 2021 | Dataset-specific design effort | Self-configuring nnU-Net pipeline | 23 public segmentation datasets | Challenge metrics (Dice-based) | Surpassed most existing, including specialised, methods | Heavy 3D training; resource-intensive |
| Work 11 [6] Wang, 2021 | Long-range context for 3D brain-tumour segmentation | 3D CNN encoder, transformer bottleneck, 3D CNN decoder | BraTS 2019 and 2020 official validation | Dice; Hausdorff distance | BraTS 2020 validation Dice: ET 0.7873, WT 0.9009, TC 0.8173 | 3D memory cost; spatial (not slice-wise) attention |
| Work 12 [7] Hatamizadeh, 2022 | Brain-tumour segmentation with hierarchical transformers | Swin transformer encoder at five resolutions with CNN decoder | BraTS 2021 challenge | Dice; HD95 | Among top-ranking methods in validation phase | High GPU memory and training time |

# Chapter 4

# Requirements

## 4.1 Functional requirements

The system is a research pipeline rather than an interactive application, so its functional requirements describe the capabilities needed to carry out and verify the study. Each requirement has an acceptance criterion that was checked during the project.

**FR1 – Data preparation.** The pipeline shall read the original BraTS 2020 NIfTI volumes directly from the downloaded archive, select the 369 annotated training cases, normalise intensities per patient and per modality over brain voxels, crop to a fixed in-plane size around the brain and store each patient as memory-mappable arrays. *Acceptance:* all 369 cases are converted, and the preprocessing log reports how many tumour voxels, if any, fall outside the crop.

**FR2 – Leakage-free splitting.** The pipeline shall split data at the patient level into five cross-validation folds, with each test fold disjoint from its training and validation patients. *Acceptance:* the stored split files contain no patient in more than one subset, and every patient appears in exactly one test fold.

**FR3 – 2.5D model.** The system shall segment the centre slice of a $K$-slice stack into WT, TC and ET masks using the cross-slice transformer, and shall reduce exactly to a 2D U-Net when $K = 1$. *Acceptance:* the same code trains both configurations by changing one argument.

**FR4 – Training with model selection.** The system shall train with validation-based checkpoint selection, resume automatically after interruption and stop safely before a scheduler time limit while still producing test results. *Acceptance:* interrupted runs resume from their last checkpoint, and every completed run writes a summary with test metrics.

**FR5 – Volumetric evaluation.** The system shall segment whole test volumes and compute per-patient 3D Dice, HD95 and false-positive-slice counts following BraTS conventions. *Acceptance:* metrics are computed per patient and averaged with equal weight per patient.

**FR6 – Hyper-parameter search.** The system shall run an Optuna search over training

hyper-parameters on validation data only, with pruning of unpromising trials. *Acceptance:* the search records every trial and reports the best configuration and parameter importances.

**FR7 – Aggregation and statistics.** The system shall aggregate results across runs and folds and perform paired per-patient significance tests between configurations. *Acceptance:* a single command produces the results table and Wilcoxon tests.

## 4.2 Non-functional requirements

**Rigour and validity.** Model comparisons must differ in exactly one factor; test patients must never influence training, model selection or post-processing choices; and claims of improvement must be supported by paired statistical tests across patients. **Reproducibility.** All experiments must be driven by versioned code and fixed random seeds, with every run's configuration, split and history saved alongside its results. **Efficiency.** The model must remain lightweight (under about 8 million parameters, comparable to the base CST paper) and train within the lab access window; a single training run should complete in about two hours on one shared A100 GPU. **Robustness of execution.** Jobs must tolerate preemption and shared-GPU conditions by checkpointing every epoch, selecting the least-loaded visible GPU and failing clearly with logs. **Privacy and ethics.** Only de-identified public data are used, no patient data are redistributed with the code, and credentials (such as the Kaggle API token) are never committed to the repository.

## 4.3 Hardware requirements

Two hardware platforms were used, as summarised in Table 4.1. The laptop was sufficient for development and for the preliminary design study but not for full-scale training, because a 4 GB GPU limited the batch size to four stacks and the per-slice HDF5 data made training input-bound at about six minutes per epoch. The final experiments therefore used the Astra cluster.

Table 4.1: Hardware used in the project

| | **Development laptop** | **Astra HPC cluster (final runs)** |
|---|---|---|
| GPU | NVIDIA RTX 3050 Laptop, 4 GB | NVIDIA A100-PCIe, 40 GB (1 per job; 6 in cluster, 3 jobs allowed) |
| CPU / memory | 16 logical cores, 16 GB RAM | 16 cores and 128 GB RAM per job |
| Driver / CUDA | CUDA 12.6 build of PyTorch | Driver 520.61.05, CUDA 11.8 |
| Storage | Local NVMe | Shared Lustre file system (`/home`) |
| Access | Local | Altair Access portal; PBS scheduler, gpuq queue |
| Role | Development, design study, unit tests | Preprocessing, Optuna search, all reported experiments |

## 4.4 Software and data requirements

The software stack is listed in Table 4.2. On Astra, the PyTorch version was pinned to 2.5.1 built against CUDA 11.8 because the cluster driver supports CUDA 11.8 at most, and all code was kept compatible with that version.

Table 4.2: Software stack

| Component | Version and purpose |
|---|---|
| Python | 3.9.15 on Astra (3.13 on the laptop) |
| PyTorch | 2.5.1+cu118 on Astra (2.7.0 on the laptop); model, bf16 mixed precision |
| NumPy, SciPy | Array processing; Euclidean distance transform for HD95; Wilcoxon signed-rank test |
| NiBabel | Reading NIfTI volumes directly from the dataset archive |
| h5py | Reading the per-slice HDF5 copy used during development |
| Optuna | Hyper-parameter search (TPE sampler, median pruner, journal storage) |
| Kaggle CLI / API | Downloading the dataset onto the cluster |
| Git and GitHub | Version control and code transfer to the cluster |
| Matplotlib, LaTeX | Figures and this report |

**Dataset.** The study uses the BraTS 2020 training set [1, 2, 14]: 369 pre-operative glioma patients from multiple institutions, each with co-registered, skull-stripped T1, T1ce, T2 and FLAIR volumes of $240 \times 240 \times 155$ voxels at 1 mm isotropic resolution, and an expert segmentation with labels 1 (necrotic and non-enhancing core), 2 (peritumoural oedema) and 4 (enhancing tumour). The original NIfTI release was obtained from a public Kaggle mirror of the challenge data (about 4.2 GB compressed) and processed directly from the archive. The BraTS validation cases in the same archive have no public masks and were not used. During development we also used a per-slice HDF5 version of the same 369 patients, in which every slice had been z-scored independently; Section 5.2.8 explains why the final pipeline returned to the original volumes.

**Preprocessing.** For each patient the four modalities are stacked; the brain mask is the set of voxels that are non-zero in any modality; each modality is z-score normalised using the mean and standard deviation of its brain voxels, with the background set to zero; axial slices containing no brain are discarded; and a $192 \times 192$ window centred on the brain's bounding box is cropped. Labels are mapped to the three nested regions WT $= \{1,2,4\}$, TC $= \{1,4\}$ and ET $= \{4\}$. Images are stored as 16-bit floating point and labels as 8-bit integers, about 44 MB per patient.

**Partitioning.** The 369 patients are shuffled once with a fixed seed (42) and divided into five folds (four of 74 patients and one of 73). For fold $f$, the patients of fold $f$ form the test set, 30 of the remaining patients form the validation set used for checkpoint selection and threshold tuning, and the other 265 (or 266) form the training set. A training set contains about 36,500

slices, of which about 17,500 contain tumour. Table 4.3 summarises the partition.

Table 4.3: Patient-level five-fold partition of the 369 BraTS 2020 training patients

| **Fold** | **0** | **1** | **2** | **3** | **4** |
|---|---|---|---|---|---|
| Training patients | 265 | 265 | 265 | 265 | 266 |
| Validation patients (checkpoint and threshold selection) | 30 | 30 | 30 | 30 | 30 |
| Test patients (used once) | 74 | 74 | 74 | 74 | 73 |
| Test patients without enhancing tumour | 4 | 6 | 10 | 4 | 3 |

# Chapter 5

# Proposed Methodology

## 5.1 Design (architecture) or conceptual framework

The proposed model, shown in Figure 5.1, is a 2.5D Cross-Slice U-Net. Its input is a stack of $K$ consecutive axial slices centred on the slice to be segmented, each slice having four channels (FLAIR, T1, T1ce and T2). A single 2D U-Net encoder, whose weights are shared across slices, processes every slice independently and produces a bottleneck feature map at one-eighth resolution together with skip-connection features at three finer scales. At the bottleneck, a cross-slice transformer treats the $K$ feature vectors found at the same spatial position in the $K$ slices as a short sequence of tokens and applies self-attention across this slice axis, so that the centre slice can draw on information from its neighbours. The fused centre-slice features are passed to a U-Net decoder, which also receives the skip connections of the centre slice only, and a final $1 \times 1$ convolution produces three logits per pixel for WT, TC and ET.

The design rests on two assumptions. First, adjacent slices of a glioma are strongly correlated, so features of neighbouring slices carry useful evidence about ambiguous regions in the centre slice. Second, this information is most useful at the bottleneck, where features are semantic and spatially compact; fusing there keeps the attention cheap (sequences of length $K$ at $24 \times 24$ positions for a $192 \times 192$ input), while the centre slice's own skip connections preserve fine boundary detail. Setting $K = 1$ removes the transformer and leaves a standard 2D U-Net, so the baseline and the proposed model share every other component.

## 5.2 Proposed approach and procedure

The complete procedure is summarised in Figure 5.2. The original MRI volumes are preprocessed once and divided into five patient-level folds. In every fold the proposed model and the baseline are trained on the same training patients. Each selects its best checkpoint and its post-processing threshold on the same validation patients, and both are then evaluated once

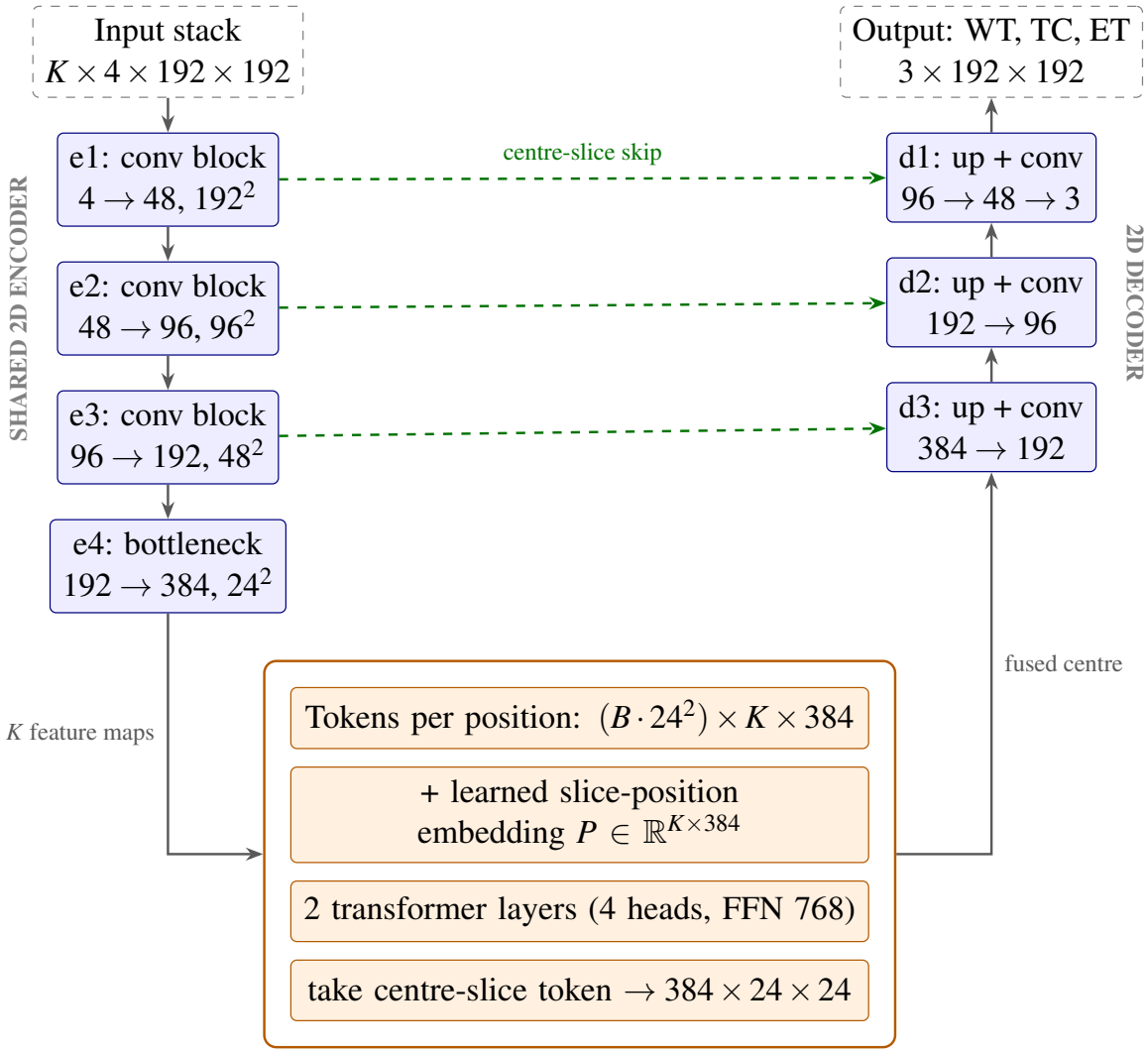



Figure 5.1: Architecture of the proposed 2.5D Cross-Slice U-Net (shapes for $K = 5$, base width 48)

on the same unseen test patients. The per-patient results of all folds are finally pooled and compared with paired statistical tests. Algorithms 1 and 2 give the training and inference steps, and the following subsections describe each stage.

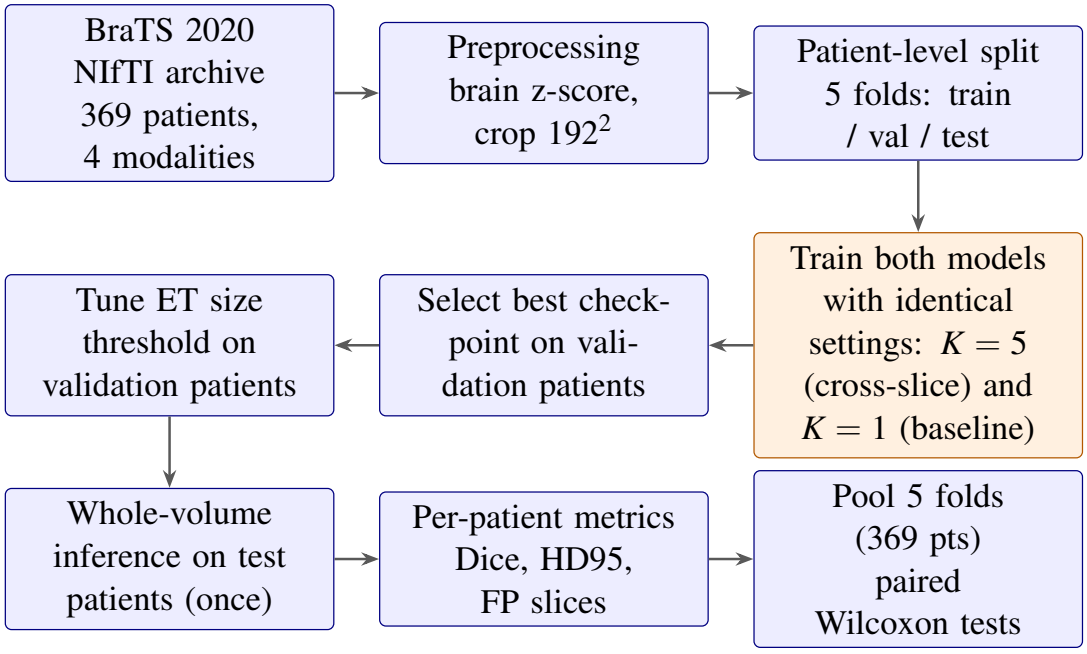



Figure 5.2: End-to-end workflow of the study, repeated for each of the five folds

### 5.2.1 2.5D input construction

For a target slice $z$ of a patient volume with $S$ slices, the input stack contains slices $z-h,\ldots,z+h$ with $h=\lfloor K/2\rfloor$. At the top and bottom of the volume, indices outside $[0,S-1]$ are clamped to the nearest valid slice, so every slice can be a target. The label of the stack is the three-region mask of slice $z$.

### 5.2.2 Network

Each conv block applies two $3\times 3$ convolutions, each followed by batch normalisation and ReLU; the encoder halves the resolution with $2\times 2$ max pooling between blocks, and the decoder doubles it with $2\times 2$ transposed convolutions before concatenating the corresponding skip features. With base width 48 the encoder produces bottleneck features $F_k\in\mathbb{R}^{384\times 24\times 24}$ for each slice $k\in\{-h,\ldots,h\}$. For every batch element and spatial position $(i,j)$, the cross-slice transformer forms the token sequence

$$X_{ij}=\big[F_{-h}(i,j);\ \ldots;\ F_h(i,j)\big]+P,\qquad X_{ij}\in\mathbb{R}^{K\times 384},\tag{5.1}$$

where $P$ is a learned slice-position embedding with one vector per slice offset, so that the network can distinguish the centre slice from slices above and below it. Two standard transformer encoder layers [27] then apply multi-head self-attention to each sequence $X_{ij}$. For

**Algorithm 1** Training one model on one fold

**Require:** training patients $\mathscr{T}$, validation patients $\mathscr{V}$, slice count $K$, epochs $E = 30$, iterations per epoch $I = 400$, batch size $B = 32$

initialise network $\theta$ and its EMA copy $\theta_{\text{EMA}} \leftarrow \theta$; best score $s^* \leftarrow -1$
**for** epoch $e = 1$ to $E$ **do**
  **for** iteration $= 1$ to $I$ **do**
    sample $B$ slices from $\mathscr{T}$, tumour slices with $5\times$ weight
    build $K$-slice stacks with edge clamping and region targets (WT, TC, ET)
    apply the same random flip, rotation and scaling to every slice of a stack; add intensity jitter
    compute loss $\mathscr{L}$ (Dice + BCE) in bfloat16; back-propagate; clip gradient norm to 1
    AdamW step with warm-up/cosine learning rate; $\theta_{\text{EMA}} \leftarrow 0.999\,\theta_{\text{EMA}} + 0.001\,\theta$
  **end for**
  **if** $e$ is even **or** $e = E$ **then**
    segment every volume in $\mathscr{V}$ with $\theta_{\text{EMA}}$; $s \leftarrow$ mean 3D Dice
    **if** $s > s^*$ **then**
      $s^* \leftarrow s$; save $\theta_{\text{EMA}}$ as the best checkpoint
    **end if**
  **end if**
**end for**
choose ET threshold $T$ that maximises mean ET Dice on $\mathscr{V}$ using the best checkpoint
**return** best checkpoint and $T$

**Algorithm 2** Segmenting and scoring one test patient

**Require:** preprocessed volume with $S$ slices, ground truth, best checkpoint, threshold $T$

**for** slice $z = 0$ to $S-1$ **do**
  stack slices $\min(\max(z+o,0),S-1)$ for $o = -\lfloor K/2 \rfloor, \ldots, \lfloor K/2 \rfloor$
  predict sigmoid probabilities for WT, TC and ET of slice $z$
**end for**
set probabilities outside the brain mask to zero; threshold at 0.5
**if** predicted ET volume $< T$ voxels **then**
  remove the predicted ET region
**end if**
compute 3D Dice and HD95 for WT, TC and ET, and the number of false-positive slices

one attention head with learned projections $W_Q$, $W_C$ and $W_V$, the queries, keys and values are $\mathbf{Q} = X_{ij}W_Q$, $\mathbf{C} = X_{ij}W_C$ and $\mathbf{V} = X_{ij}W_V$, and

$$\text{Attention}(\mathbf{Q},\mathbf{C},\mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{C}^\top}{\sqrt{d_h}}\right)\mathbf{V}, \tag{5.2}$$

where $d_h = 384/4 = 96$ is the dimension of each head. The keys are written $\mathbf{C}$ here to avoid confusion with $K$, which throughout this report denotes the number of input slices; the $K \times K$ attention matrix gives, for every slice, how strongly it draws on each of the other slices. The layers use four heads, a feed-forward width of 768 and dropout 0.1, followed by residual connections and layer normalisation. Only the output token of the centre slice is kept, reshaped into a $384 \times 24 \times 24$ map and decoded. The proposed model has 6.70 M parameters, of which the cross-slice transformer contributes 2.37 M; the $K = 1$ baseline has 4.33 M.

### 5.2.3 Training objective and sampling

The network is trained directly on the three overlapping evaluation regions with independent sigmoid outputs. With predicted probabilities $p$, binary targets $t$, regions $r \in \{\text{WT}, \text{TC}, \text{ET}\}$ and smoothing constant $s = 1$, the loss combines soft Dice and binary cross-entropy:

$$\mathscr{L} = (1-\lambda)\left[1 - \frac{1}{3}\sum_r \frac{2\sum p_r t_r + s}{\sum p_r + \sum t_r + s}\right] + \lambda\,\text{BCE}(p,t), \qquad \lambda = 0.5. \tag{5.3}$$

All slices, including those without tumour, are used for training, so that the model learns to predict empty masks on healthy tissue. Because only about 48% of slices contain tumour and tumour pixels are a small fraction of each slice, tumour-containing slices are drawn five times as often as other slices by weighted random sampling, following the oversampling strategy of the base paper [8].

### 5.2.4 Augmentation

Augmentation is applied on the GPU to whole batches. The same geometric transformation is applied to all $K$ slices of a stack and to its mask, so the alignment between slices is preserved: random left–right and up–down flips (probability 0.5 each), a random rotation of up to $\pm 15^\circ$ combined with isotropic scaling between 0.85 and 1.15 (probability 0.5), and, on brain voxels only, a random per-modality intensity scaling in $[0.9, 1.1]$ and shift in $[-0.1, 0.1]$ plus Gaussian noise with standard deviation 0.05 (probability 0.5).

### 5.2.5 Optimisation

The model is optimised with AdamW [28] (learning rate $2 \times 10^{-4}$, weight decay $10^{-5}$) using batches of 32 stacks. Training lasts 30 epochs of 400 iterations, that is, about 384,000

sampled stacks or roughly ten passes over the training slices. The learning rate rises linearly during the first three epochs and then follows a cosine decay to $10^{-6}$ at the end of training. Gradients are clipped to a norm of 1.0, computation uses bfloat16 mixed precision, and an exponential moving average (EMA) of the weights, $\theta_{\text{EMA}} \leftarrow 0.999\,\theta_{\text{EMA}} + 0.001\,\theta$, is maintained and used for validation and testing. Every two epochs the EMA model segments all 30 validation volumes, and the checkpoint with the highest validation mean 3D Dice is kept.

### 5.2.6 Inference and post-processing

At test time each patient volume is segmented slice by slice using the same clamped $K$-slice stacks, the sigmoid outputs are restricted to the brain mask, and each region is thresholded at 0.5. Some patients have no enhancing tumour, and in such cases even a few falsely predicted ET voxels give an ET Dice of zero. Following common BraTS practice, a predicted ET region smaller than a threshold $T$ voxels is therefore removed. $T$ is chosen per run from $\{0, 50, 100, 200, 300, 500, 750, 1000\}$ to maximise the mean ET Dice on the validation patients, and it is then applied unchanged to the test patients.

### 5.2.7 Hyper-parameter justification

The training hyper-parameters were justified in two stages. In the first stage, during the mid-review phase, a one-factor-at-a-time design study varied one choice at a time around a reference configuration on a reduced setting (16 training volumes, 10 epochs, the per-slice evaluation used at the time) that the laptop GPU could handle. Its results, summarised in Table 5.1, supported $K = 5$ over $K = 3$ and $K = 1$, AdamW, the combined Dice and BCE loss and a learning rate of $2 \times 10^{-4}$. The scheduler comparison was inconclusive at that short budget, and plain SGD with the same learning rate collapsed but matched AdamW once its learning rate was tuned. Because these numbers come from four validation volumes and an earlier metric, they are treated as preliminary evidence only.

Table 5.1: Preliminary one-factor-at-a-time design study (mid-review protocol: 16 training volumes, 10 epochs, per-slice validation Dice)

| Factor | Setting | Mean Dice | Factor | Setting | Mean Dice |
|---|---|---|---|---|---|
| Slices $K$ | 1 | 0.689 | Loss | Dice + BCE (chosen) | 0.716 |
| | 3 | 0.705 | | Dice only | 0.706 |
| | 5 (chosen) | 0.716 | | BCE only | 0.690 |
| Optimiser | AdamW (chosen) | 0.716 | Learning rate | $10^{-3}$ | 0.694 |
| | Adam | 0.707 | | $2 \times 10^{-4}$ (chosen) | 0.716 |
| | SGD, lr $2 \times 10^{-4}$ | 0.221 | | $10^{-4}$ | 0.677 |
| | SGD, lr $10^{-2}$ | 0.717 | Scheduler | cosine / none / step | 0.716 / 0.724 / 0.656 |

In the second stage, following our guide's recommendation, an automated search with Optuna [10] was run on the Astra cluster. The search used a Tree-structured Parzen Estimator

sampler and a median pruner that stops trials performing below the median of earlier trials after the first four epochs. The search space covered the learning rate ($5 \times 10^{-5}$ to $10^{-3}$, log scale), weight decay ($10^{-6}$ to $10^{-3}$, log scale), BCE weight $\lambda$ (0.2–0.8), tumour-slice oversampling factor (1–10), number of slices $K \in \{3,5,7\}$, number of transformer layers (1–3) and base width $\in \{32,48\}$. Each trial trained for 8 short epochs of 200 iterations on fold 0, and its score was the validation mean 3D Dice; test patients were never used. Two trials ran concurrently on one A100 for about one hour. The results are reported in Section 6.3.4.

### 5.2.8 Evolution from the mid-review prototype

The mid-review prototype (v1) was developed under the 4 GB laptop constraint. Before moving to the cluster we audited it and re-evaluated its saved checkpoints with the standard BraTS protocol. The audit found that v1's metric computed Dice per slice, only on slices containing tumour, and counted a region absent from both prediction and ground truth as a perfect score. This understated WT and overstated TC and ET: v1 reported a mean Dice of 0.795 for the baseline, but the volumetric per-patient Dice of the same checkpoint was 0.831 (WT 0.881, TC 0.848, ET 0.764). The re-evaluation also showed that the v1 cross-slice model, although tied in Dice, produced 8.7 slices per patient with false-positive tumour against 17.7 for the baseline and roughly halved the WT boundary error (HD95 9.2 mm against 17.9 mm). This was the first evidence that the benefit of cross-slice context lies mainly in volumetric consistency.

Table 5.2 lists the issues found and the corresponding changes made for the final system (v2). The two most consequential were the evaluation protocol and the training data: v1 trained and validated only on tumour-containing slices, so the model never learned to predict empty masks on healthy tissue, and v1's per-slice HDF5 data had been normalised slice by slice, so neighbouring slices had inconsistent intensity scales, which directly undermines cross-slice fusion. v1 also used the same patients for checkpoint selection and for reporting, had no test split, and added a single shared position vector to all slice tokens, which prevented the transformer from distinguishing slice order.

### 5.2.9 Implementation on the Astra HPC cluster

The code was organised as a Python package (`cst2`) with modules for preprocessing, data loading, augmentation, the model, the loss, evaluation, training and hyper-parameter search, plus a set of cluster utilities. Jobs are submitted through the Altair Access portal as PBS jobs on the `gpuq` queue, each with one A100. We found that the portal could not launch wrapper scripts beginning with a `#!/usr/bin/env python3` line, while plain Python files without that line ran correctly, so the wrapper generator writes such files with absolute paths. Because both GPUs of a node were visible to every job, the job runner selects the least-loaded GPU before starting. Each wrapper runs a predefined *plan* of several trainings concurrently

Table 5.2: Issues found in the mid-review prototype (v1) and changes in the final system (v2)

| **Aspect** | **Mid-review prototype (v1)** | **Final system (v2)** |
|---|---|---|
| Evaluation metric | Per-slice Dice on tumour slices; empty–empty counted as 1 | Per-patient 3D Dice, HD95 and false-positive slices (BraTS rules) |
| Data splits | Train/validation only; validation also used for reporting | Patient-level 5-fold CV with separate validation and test sets |
| Training slices | Tumour-containing slices only | All slices, tumour slices oversampled 5× |
| Intensity normalisation | Per-slice z-score including background (pre-made HDF5) | Per-patient, per-modality z-score over brain voxels (original NIfTI) |
| Slice position encoding | One shared vector for all slices | One learned vector per slice offset |
| Targets | Raw classes (NCR/NET, ED, ET) | Evaluation regions (WT, TC, ET) |
| Training recipe | Batch 4, flips only, fp16, no EMA | Batch 32, geometric and intensity augmentation, bf16, warm-up, clipping, EMA |
| Post-processing | None | Validation-tuned small-ET removal |
| Scale of evaluation | One split, single run | Five folds, all 369 patients, paired tests |

on its GPU. The concurrency is set from the measured memory footprint (about 12.5 GB for $K = 5$ and 3.8 GB for $K = 1$ at batch 32), after an early configuration of four trainings per GPU exceeded the 40 GB limit. Every run checkpoints each epoch, resumes automatically if resubmitted, and stops at a configurable deadline while still performing its final test evaluation, so that results were guaranteed within the lab window.

# Chapter 6

# Results and Analysis

## 6.1 Evaluation setup

The main experiment compares the proposed cross-slice model ($K = 5$) with the 2D U-Net baseline ($K = 1$) using five-fold patient-level cross-validation over all 369 BraTS 2020 training patients. In each fold, both models are trained on the same 265 or 266 training patients, select their checkpoint on the same 30 validation patients, and are evaluated once on the same 73 or 74 test patients, so that every patient is a test patient exactly once and is segmented by both models. The two configurations share the training recipe of Chapter 5 (AdamW, learning rate $2 \times 10^{-4}$, Dice+BCE with $\lambda = 0.5$, five-fold tumour oversampling, base width 48, two transformer layers for $K = 5$, batch 32, 30 epochs of 400 iterations, cosine schedule, EMA) and the same seed; the only difference is the number of input slices and, with it, the presence of the cross-slice transformer. All runs were executed on Astra A100 GPUs, with two or three runs sharing a GPU. In every run validation never triggered early stopping, so both models received exactly the same training budget of 30 epochs.

Several safeguards protect the validity of the comparison. Splitting is by patient, so slices of one patient never appear in two subsets. The test patients are used only once, after training; checkpoint selection and the ET post-processing threshold are chosen on validation patients. Metrics are computed per patient on whole volumes, which prevents slice-level averaging from hiding errors on healthy slices. Significance is assessed with paired tests on the same patients rather than by comparing means of independent samples. The training settings of the main experiment were fixed before it was run, taken from the mid-review design study, and confirmed as near-optimal by the Optuna search (Section 6.3.4).

## 6.2 Evaluation criteria

**Dice similarity coefficient.** For a region with predicted voxel set $P$ and ground-truth set $G$ in one patient's volume,

$$\text{Dice} = \frac{2\,|P \cap G|}{|P| + |G|}, \tag{6.1}$$

which ranges from 0 (no overlap) to 1 (perfect overlap). Following the BraTS convention, if both $P$ and $G$ are empty the Dice is 1, and if exactly one is empty it is 0. Dice is computed separately for WT, TC and ET, and the mean Dice is their average. It is the primary metric.

**95th-percentile Hausdorff distance (HD95).** With $\partial P$ and $\partial G$ the surface voxels of the two regions, HD95 is the 95th percentile of the set of distances from each surface voxel of one region to the nearest surface voxel of the other, taken in both directions. It measures boundary error in millimetres (voxels are 1 mm isotropic); lower is better. If exactly one region is empty, the BraTS penalty of 373.13 mm is used, and if both are empty the distance is 0.

**False-positive slices.** The number of axial slices in a patient's volume on which tumour (WT) is predicted although the ground truth contains no tumour. It directly measures the slice-to-slice consistency that cross-slice context is meant to improve.

**Aggregation and statistics.** Each metric is computed per patient and averaged with equal weight per patient. We report the mean and standard deviation of the five fold-level averages, and the pooled per-patient mean over all 369 patients. Differences between $K = 5$ and $K = 1$ are tested with the two-sided Wilcoxon signed-rank test on the 369 paired per-patient values, which does not assume normally distributed differences. We also report the percentage of patients for which $K = 5$ is strictly better.

## 6.3 Result

### 6.3.1 Main five-fold comparison

Table 6.1 summarises the main result. Averaged over the five folds, the cross-slice model improves the Dice of all three regions and reduces the boundary error of all three regions as well as the number of false-positive slices. Figure 6.1 shows that the mean Dice of $K = 5$ is higher than that of $K = 1$ in four of the five folds, the exception being fold 1 (0.818 against 0.824). The WT Dice is higher in all five folds. The per-fold Dice values and WT boundary errors are listed in Table 6.2, and the remaining boundary and consistency metrics in Table 6.3.

Table 6.3 shows that the reduction in false-positive slices is not uniform. The cross-slice model has fewer false-positive slices in folds 2, 3 and 4 (by 1.5 to 2.2 slices per patient), but slightly more in folds 0 and 1. The ET boundary error is lower in every fold, and the TC

Table 6.1: Five-fold cross-validation results on all 369 patients (mean ± standard deviation of the five fold averages; ↑ higher is better, ↓ lower is better)

| **Metric** | **Baseline 2D U-Net** ($K = 1$) | **Cross-slice transformer** ($K = 5$) | **Folds improved** |
|---|---|---|---|
| WT Dice ↑ | $0.898 \pm 0.010$ | $\mathbf{0.904 \pm 0.006}$ | 5 / 5 |
| TC Dice ↑ | $0.837 \pm 0.021$ | $\mathbf{0.840 \pm 0.028}$ | 2 / 5 |
| ET Dice ↑ | $0.783 \pm 0.042$ | $\mathbf{0.789 \pm 0.041}$ | 4 / 5 |
| Mean Dice ↑ | $0.839 \pm 0.020$ | $\mathbf{0.844 \pm 0.022}$ | 4 / 5 |
| HD95 WT (mm) ↓ | $7.7 \pm 1.0$ | $\mathbf{5.9 \pm 0.6}$ | 4 / 5 |
| HD95 TC (mm) ↓ | $9.6 \pm 4.5$ | $\mathbf{7.9 \pm 4.9}$ | 4 / 5 |
| HD95 ET (mm) ↓ | $33.4 \pm 19.0$ | $\mathbf{28.3 \pm 15.6}$ | 5 / 5 |
| False-positive slices ↓ | $2.46 \pm 1.04$ | $\mathbf{1.62 \pm 0.43}$ | 3 / 5 |
| Parameters | 4.33 M | 6.70 M | – |
| Training time per run (A100, shared) | 0.7–1.0 h | 1.3–2.0 h | – |

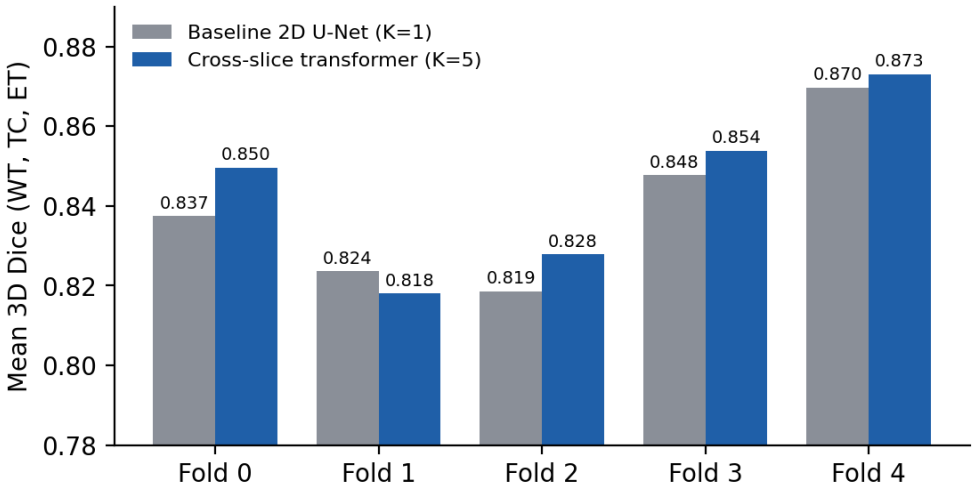



Figure 6.1: Mean 3D Dice per cross-validation fold for the baseline and the proposed model

Table 6.2: Per-fold test results (73–74 patients per fold)

| | **Baseline** ($K = 1$) | | | | **Proposed** ($K = 5$) | | | |
|---|---|---|---|---|---|---|---|---|
| **Fold** | WT | TC | ET | HD95 WT | WT | TC | ET | HD95 WT |
| 0 | 0.891 | 0.849 | 0.772 | 7.2 | 0.905 | 0.867 | 0.777 | 6.5 |
| 1 | 0.884 | 0.810 | 0.776 | 6.5 | 0.894 | 0.807 | 0.753 | 6.5 |
| 2 | 0.903 | 0.826 | 0.727 | 8.5 | 0.907 | 0.824 | 0.753 | 5.8 |
| 3 | 0.908 | 0.835 | 0.800 | 9.0 | 0.910 | 0.830 | 0.821 | 5.6 |
| 4 | 0.902 | 0.866 | 0.841 | 7.4 | 0.905 | 0.872 | 0.842 | 5.1 |

Table 6.3: Per-fold boundary error for TC and ET and false-positive slices per patient

| | **Baseline** ($K = 1$) | | | **Proposed** ($K = 5$) | | |
|---|---|---|---|---|---|---|
| **Fold** | HD95 TC (mm) | HD95 ET (mm) | FP slices | HD95 TC (mm) | HD95 ET (mm) | FP slices |
| 0 | 8.4 | 29.4 | 1.30 | 6.0 | 28.2 | 2.34 |
| 1 | 16.3 | 39.2 | 1.41 | 16.5 | 30.1 | 1.66 |
| 2 | 11.8 | 62.8 | 3.04 | 7.0 | 53.1 | 1.26 |
| 3 | 5.9 | 22.6 | 2.97 | 5.4 | 17.4 | 1.43 |
| 4 | 5.5 | 12.9 | 3.58 | 4.8 | 12.8 | 1.41 |
| Mean | 9.6 | 33.4 | 2.46 | 7.9 | 28.3 | 1.62 |

boundary error in four of the five folds.

### 6.3.2 Paired per-patient analysis

Table 6.4 pools all 369 patients and compares the two models patient by patient. All Dice differences favour the cross-slice model and are statistically significant ($p < 10^{-5}$), although they are small in magnitude: $+0.006$ for WT and ET and $+0.003$ for TC. The proposed model achieves a higher WT Dice for 71.5% of patients and a higher mean Dice for 68.8% of patients. The boundary metrics show larger relative gains: the mean HD95 decreases by 1.8 mm for WT (23% lower), 1.6 mm for TC and 5.1 mm for ET (15% lower), and the mean number of false-positive slices per patient decreases from 2.46 to 1.62 (34% lower), all significant at the 5% level. Figure 6.2 shows the distribution of the per-patient differences. For most patients the differences are close to zero, and the improvement in the means comes from a consistent small shift in Dice together with the removal of large boundary errors in a minority of difficult cases. This explains why the median HD95 changes little while the mean falls substantially. Figure 6.3 shows the fold-level HD95 for each region: the cross-slice model has a lower mean boundary error for all three regions, and for WT its fold-to-fold spread is also smaller (standard deviation 0.6 mm against 1.0 mm), indicating more stable boundaries across patient groups. The large ET values and spread in both models come from patients without enhancing tumour: 27 of the 369 patients have no ET, ten of them in fold 2, and in such a case a single spurious ET prediction receives the BraTS penalty distance of 373.13 mm. Across all folds this penalty was incurred for 30 patients by the baseline and for 25 by the cross-slice model, which accounts for much of the reduction in mean ET HD95.

### 6.3.3 Training behaviour

Figure 6.4 shows the training loss and the validation mean 3D Dice for fold 0. Both models converge smoothly. The baseline reached its best validation score at epoch 16, while the cross-slice model continued to improve until epoch 28, as the learning rate approached zero. In folds 1–4 the best validation epoch of the cross-slice model was between 14 and 18 and that of the baseline between 14 and 16. Training a $K = 5$ model took 1.3–2.0 hours and

Table 6.4: Paired per-patient comparison over all 369 patients (difference = $K = 5$ minus $K = 1$; two-sided Wilcoxon signed-rank test)

| **Metric** | $K = 1$ **mean** | $K = 5$ **mean** | **Mean difference** | **Patients improved** | $p$**-value** |
|---|---|---|---|---|---|
| WT Dice | 0.898 | 0.904 | +0.006 | 71.5% | $7 \times 10^{-18}$ |
| TC Dice | 0.837 | 0.840 | +0.003 | 63.1% | $2 \times 10^{-6}$ |
| ET Dice | 0.783 | 0.789 | +0.006 | 55.6% | $3 \times 10^{-6}$ |
| Mean Dice | 0.839 | 0.844 | +0.005 | 68.8% | $8 \times 10^{-11}$ |
| HD95 WT (mm) | 7.71 | 5.91 | −1.80 | 53.7% | $2 \times 10^{-11}$ |
| HD95 TC (mm) | 9.57 | 7.92 | −1.65 | 39.8% | 0.032 |
| HD95 ET (mm) | 33.46 | 28.39 | −5.07 | 24.7% | 0.016 |
| False-positive slices | 2.46 | 1.62 | −0.83 | 33.6% | $2 \times 10^{-7}$ |

"Patients improved" counts strict improvements; for HD95 and false-positive slices many patients tie (for example, zero false-positive slices for both models).

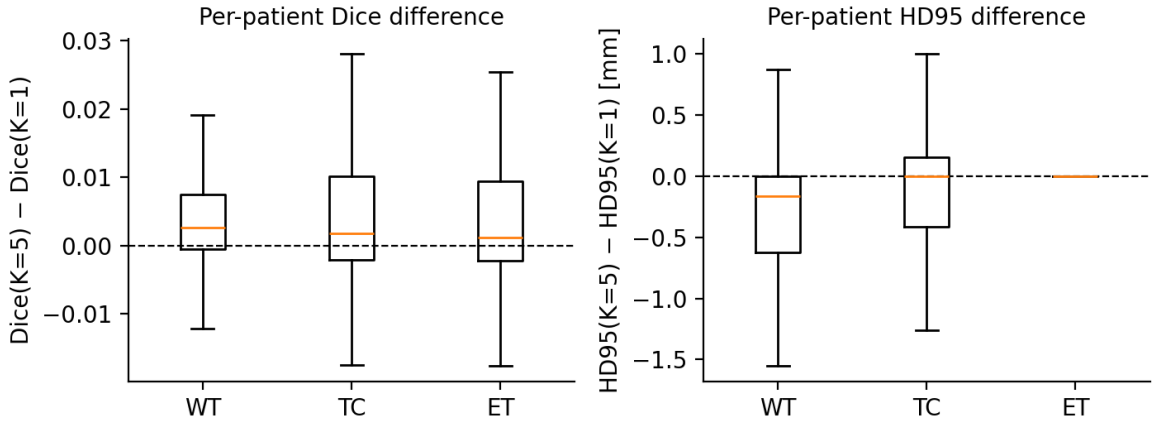



Figure 6.2: Distribution of per-patient differences between the proposed model and the baseline over 369 patients (boxes show the interquartile range; outliers omitted)

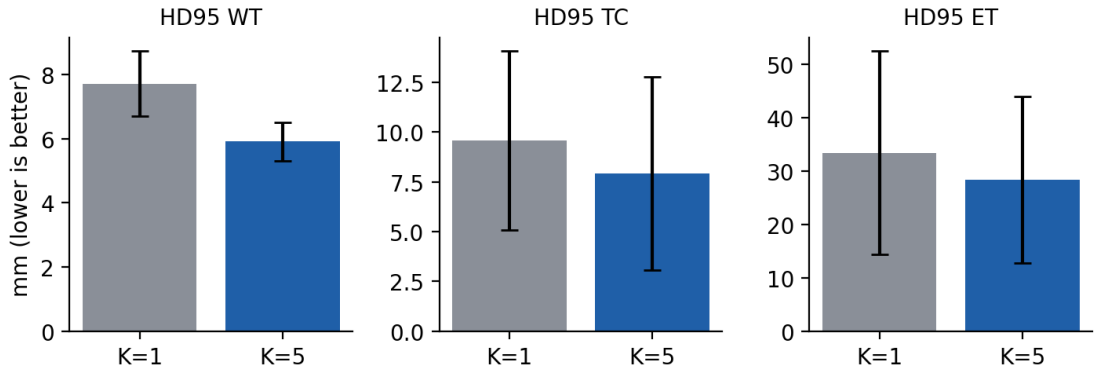



Figure 6.3: HD95 boundary error per region (mean and standard deviation over the five folds)

a $K = 1$ model 0.7–1.0 hours on a shared A100, with peak GPU memory of 12.5 GB and 3.8 GB respectively. The ET size threshold selected on validation patients ranged from 0 to 1,000 voxels across runs.

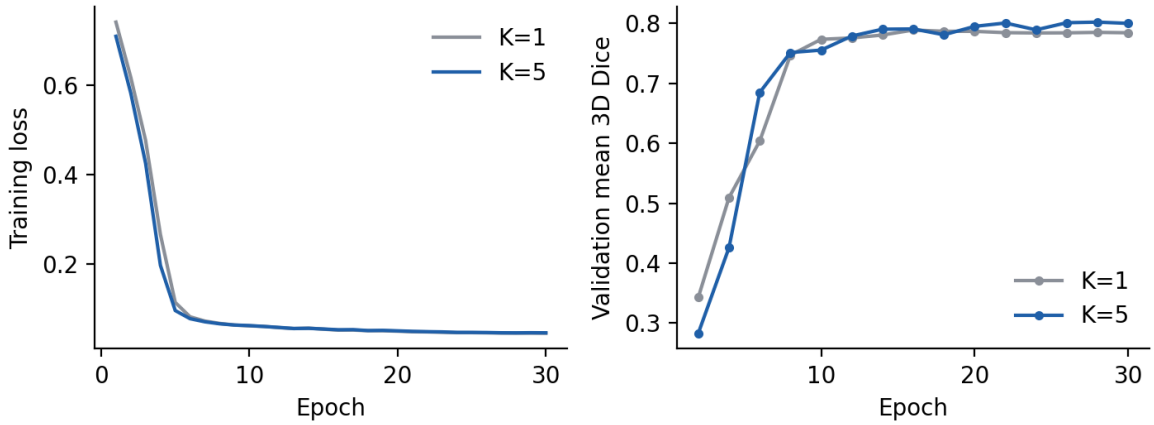



Figure 6.4: Training loss and validation mean 3D Dice per epoch on fold 0

### 6.3.4 Automated hyper-parameter search

The Optuna search ran ten trials in its one-hour budget: six completed and four were pruned early. Figure 6.5 and Table 6.5 summarise them. The best trial used $K = 5$, a learning rate of $1.76 \times 10^{-4}$, weight decay $1.1 \times 10^{-4}$, BCE weight 0.32, tumour oversampling 8.9, two transformer layers and base width 48, and reached a validation Dice of 0.757 in its short budget. Its slice count, learning rate, depth and width match the configuration chosen in the mid-review design study ($K = 5$, $2 \times 10^{-4}$, two layers, width 48). Trials with $K = 7$ and $K = 3$ scored 0.736 and 0.711. The fANOVA importance analysis attributed most of the variation in validation Dice to the BCE weight (32%), the learning rate (29%), the oversampling factor (23%) and the weight decay (14%). With only ten trials, these importances and the apparent zero importance of $K$ and width are not reliable estimates; the search is best read as an independent confirmation that the hand-chosen configuration lies in a good region of the search space.

Table 6.5: Completed Optuna trials (validation mean 3D Dice after 8 short epochs, fold 0)

| Trial | $K$ | Learning rate | Weight decay | BCE weight | Oversampling | Layers / width | Val. Dice |
|---|---|---|---|---|---|---|---|
| 3 | 5 | $1.8 \times 10^{-4}$ | $1.1 \times 10^{-4}$ | 0.32 | 8.9 | 2 / 48 | **0.757** |
| 0 | 7 | $1.7 \times 10^{-4}$ | $1.4 \times 10^{-4}$ | 0.20 | 3.7 | 2 / 48 | 0.736 |
| 7 | 3 | $6.7 \times 10^{-5}$ | $1.8 \times 10^{-5}$ | 0.77 | 5.8 | 3 / 48 | 0.711 |
| 2 | 5 | $5.4 \times 10^{-4}$ | $3.9 \times 10^{-5}$ | 0.54 | 9.3 | 3 / 48 | 0.703 |
| 4 | 5 | $9.4 \times 10^{-4}$ | $2.5 \times 10^{-4}$ | 0.48 | 8.0 | 3 / 32 | 0.677 |
| 1 | 5 | $2.6 \times 10^{-4}$ | $1.4 \times 10^{-4}$ | 0.56 | 5.9 | 3 / 32 | 0.675 |

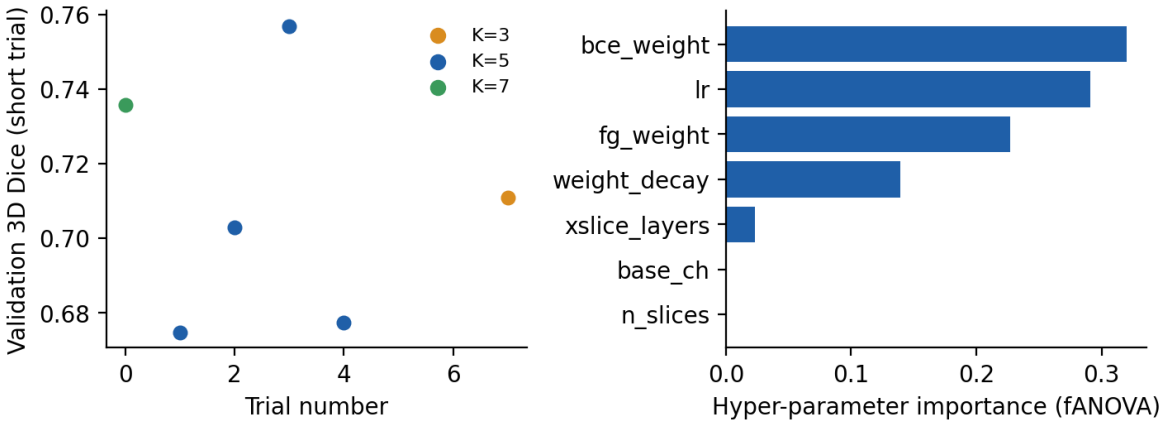



Figure 6.5: Optuna search: validation Dice of completed trials coloured by slice count $K$ (left) and hyper-parameter importance (right)

### 6.3.5 Supplementary experiment: long training with early stopping

Before the main experiment we ran a second protocol overnight. It used the best Optuna settings (Section 6.3.4), a 150-epoch cosine learning-rate schedule, validation every four epochs and early stopping after ten validation checks (40 epochs) without improvement, so that each run stops 40 epochs after its best validation epoch. Two of the five $K = 5$ runs (folds 0 and 2) failed with GPU out-of-memory errors, because four trainings shared one 40 GB GPU, so a paired comparison is available for folds 1, 3 and 4 only. Table 6.6 lists, for each of these folds, the test mean Dice, the epoch with the best validation score, the epoch at which training stopped and the learning rate at that point.

Table 6.6: Supplementary long-training experiment with early stopping (folds in which both runs completed; training stops 40 epochs after the best validation epoch)

| | **Baseline** ($K = 1$) | | | | **Proposed** ($K = 5$) | | | |
|---|---|---|---|---|---|---|---|---|
| **Fold** | Mean Dice | Best epoch | Stopped at | LR at stop | Mean Dice | Best epoch | Stopped at | LR at stop |
| 1 | **0.835** | 56 | 96 | $5.3 \times 10^{-5}$ | 0.819 | 16 | 56 | $1.3 \times 10^{-4}$ |
| 3 | **0.850** | 48 | 88 | $6.7 \times 10^{-5}$ | 0.848 | 12 | 52 | $1.3 \times 10^{-4}$ |
| 4 | **0.874** | 40 | 80 | $8.2 \times 10^{-5}$ | 0.868 | 92 | 132 | $7.4 \times 10^{-6}$ |

In this protocol the baseline scored higher in all three folds, but the differences are small: pooled over the 221 patients of these folds, none of the paired Dice differences is statistically significant (WT $+0.000$, TC $-0.009$, ET $-0.015$; all $p > 0.09$). The number of false-positive slices was higher for the cross-slice model ($+0.85$ per patient, $p = 0.001$), the opposite of the main experiment.

The table shows that the two models did not receive comparable training in every fold. In folds 1 and 3 the cross-slice model reached its best validation score early (epochs 16 and 12) and was stopped at epochs 56 and 52, while the learning rate was still about two-thirds of its maximum. The baseline trained until epochs 96 and 88. In these two folds the cross-slice

model therefore never entered the low-learning-rate phase, in which it gained most in the main experiment. Fold 4, however, does not follow this pattern: there the cross-slice model trained longer than the baseline (132 against 80 epochs, with its learning rate almost fully annealed) and still scored 0.006 lower. Unequal training therefore explains part, but not all, of the outcome of this experiment.

Two further differences from the main protocol may contribute. The Optuna settings were selected from eight-epoch trials, which favour configurations that converge quickly rather than those that generalise best after long training; compared with the main protocol they used a lower BCE weight (0.32 against 0.5) and a higher tumour oversampling (8.9 against 5), both of which shift the balance between false positives and false negatives. In addition, with only 30 validation patients the early-stopping decision is noisy, so the selected checkpoint can come from a local peak of the validation curve. The experiment therefore shows that the advantage of cross-slice context is not robust to every training protocol. When the two models were trained under the main protocol (identical fixed length, fully annealed schedule and the design-study settings), the cross-slice model was better on all 369 patients; under this long early-stopped protocol with short-trial tuned settings, the two models were statistically indistinguishable. Comparing the protocols also shows that the extra training gave the baseline only a small benefit (mean Dice $+0.005$ on average over its five folds relative to the 30-epoch runs), which indicates that 30 annealed epochs are close to convergence for the baseline. We report this experiment in full because it limits how strongly the main result can be generalised; Section 6.6 discusses this further.

## 6.4 Comparative Analysis

**Against the matched baseline.** The controlled comparison above is the central comparative result. Under identical data, splits, training budget and settings, adding cross-slice attention improves all three Dice scores by a small but statistically reliable margin and reduces boundary error and false-positive slices more substantially, at a cost of 2.37 M additional parameters and roughly 1.6–2 times the training time.

**Against the mid-review prototype.** With the corrected pipeline the absolute performance of both models improved. The v1 checkpoints, re-evaluated volumetrically on their own validation split, reached a mean Dice of 0.831 (baseline) and 0.826 (cross-slice). The final system reaches 0.839 and 0.844 averaged over all 369 patients in cross-validation. More importantly, the relative result changed: v1's full-data comparison was a tie in Dice, whereas the corrected protocol shows a consistent, significant advantage for cross-slice context.

**Against published methods.** Table 6.7 places our results beside published BraTS 2020 results for orientation. The numbers are *not* directly comparable. Published results such as TransBTS were obtained on the official BraTS validation set through the challenge server,

whereas ours come from internal five-fold cross-validation on the training set, and the MM-MSCA-AF "overall" Dice is not broken down by region in its abstract. Our results nevertheless indicate that a lightweight 2.5D model with fewer than 7 M parameters reaches the same range as recent 3D and hybrid methods on the BraTS regions.

Table 6.7: Context against published BraTS 2020 results (different evaluation sets; for orientation only)

| **Method** | **Evaluation** | **WT** | **TC** | **ET** |
|---|---|---|---|---|
| TransBTS (3D CNN + transformer) [6] | Official validation set | 0.901 | 0.817 | 0.787 |
| MM-MSCA-AF [11] | BraTS 2020 (overall Dice 0.859) | – | – | – |
| Baseline 2D U-Net (ours) | 5-fold CV, 369 patients | 0.898 | 0.837 | 0.783 |
| Cross-slice transformer (ours) | 5-fold CV, 369 patients | 0.904 | 0.840 | 0.789 |

**Against the base paper.** Abdikenov *et al.* reported that adaptive attention added $+0.006$ Dice in CT lymph-node segmentation [8]. Our adaptation to MRI yields a Dice gain of the same order ($+0.005$ mean, $+0.006$ for WT and ET). It also shows clearer improvements in boundary accuracy and slice consistency, which the base paper did not measure in the same way.

## 6.5 Discussion and Interpretation

The results answer the research question in a nuanced but positive way. Cross-slice attention does make 2.5D glioma segmentation more accurate than an otherwise identical 2D network, and the improvement is consistent across patients and folds. Its size in Dice, however, is modest, about half a Dice point. This is plausible: Dice is dominated by the bulk of large tumours, which a strong 2D network already segments well from a single slice. Neighbouring slices matter most where the evidence in the centre slice is ambiguous, near the top and bottom of the tumour, at thin boundaries and in small enhancing regions. Errors there have little effect on overlap but a large effect on boundary distance and on whether spurious tumour appears on a slice. This is exactly where the cross-slice model gains most: 23% lower WT boundary error, 15% lower ET boundary error and a third fewer slices with false-positive tumour.

These findings support the motivation of the project. The objective was not only higher overlap but segmentations that are consistent through the volume, and the HD95 and false-positive-slice results provide direct, statistically tested evidence for that property. For the intended application, consistent boundaries and fewer spurious detections on healthy slices are what reduce a clinician's correction effort, so this kind of improvement matters more in practice than its Dice value suggests. The cost is moderate: the model remains under 7 M parameters and trains in about two hours on one GPU, far below the memory and time

demands of the 3D methods in Chapter 3.

The project also produced a methodological lesson. Under the mid-review protocol the comparison appeared to be a tie on full data, and in the supplementary long-training experiment the two models were again statistically indistinguishable. In that experiment early stopping gave the models unequal training in two of three folds, but not in the third. Only when the evaluation followed BraTS conventions, the data were normalised per patient, and both models received identical, fully annealed training did a consistent difference emerge. Small architectural gains are therefore very sensitive to the evaluation and training protocol, and the claim supported by this work is correspondingly specific: under a controlled, equal-budget protocol, cross-slice context yields a small but consistent Dice gain and a larger improvement in boundary accuracy and slice consistency. This is why we report both experiments and base our claims on paired tests over all patients rather than on a single favourable run.

## 6.6 Limitations

Several limitations qualify these findings. First, each configuration was trained once per fold with a single seed, so seed-to-seed variance is not quantified, and the Dice improvement (about 0.005) is comparable to typical seed variation in segmentation networks. The paired test over 369 patients establishes consistency across patients for the trained models but not robustness to retraining. Second, performance was estimated by internal cross-validation on the BraTS 2020 training set; results on the official validation set, other BraTS editions or external clinical data were not measured. Third, the advantage is protocol-dependent: under the supplementary long-training protocol with early stopping and short-trial Optuna settings it was not observed (Section 6.3.5), and the reasons are only partly explained. Fourth, the Optuna search was short (ten trials, one hour), so its importance estimates are unreliable. Fifth, the cross-slice module attends over learned bottleneck features. This is a deliberate variant of the base paper's descriptor-based fusion of raw slices, and the paper-faithful variant was not implemented for comparison. Sixth, the gain in TC Dice is small and inconsistent across folds (two of five), and fold 1 shows a small loss in mean Dice. Seventh, no qualitative visual assessment by clinicians, no 3D reference model trained under the same conditions and no measurement of inference time in a clinical workflow were carried out. Finally, the work uses a public research dataset and makes no claim of clinical validity.

# Chapter 7

# Conclusion

## 7.1 Conclusion

This project addressed the inconsistency of slice-wise automated glioma segmentation by adapting the cross-slice transformer idea from CT lymph-node segmentation to multi-modal brain MRI. We designed and implemented a 2.5D Cross-Slice U-Net that encodes five neighbouring slices with a shared encoder and fuses them through self-attention across the slice axis at the bottleneck. The same code reduces to a 2D U-Net baseline when $K = 1$. We replaced the mid-review prototype's evaluation and data handling with a rigorous pipeline: original NIfTI data normalised per patient, patient-level five-fold cross-validation with separate validation and test sets, training on all slices, and volumetric per-patient metrics with paired statistical tests. We also justified the hyper-parameters with a design study and an Optuna search, and ran the full-scale experiments on the institutional Astra A100 cluster.

The objectives were achieved as follows. The model was implemented (objective 1), and the leakage-free, reproducible pipeline is publicly available (objective 2). The controlled five-fold comparison on all 369 BraTS 2020 patients showed that cross-slice context improves Dice for all three regions (WT 0.904 vs 0.898, TC 0.840 vs 0.837, ET 0.789 vs 0.783), with small but statistically significant gains (objective 3). The hyper-parameters were justified, and the Optuna search independently confirmed the chosen configuration (objective 4). Boundary error and slice consistency improved markedly: HD95 fell by 1.8 mm for WT and 5.1 mm for ET, and false-positive slices fell by 34% (objective 5). The full experiments were completed on the HPC cluster (objective 6). These findings are consistent with the results discussed in Chapter 6 and support the conclusion that cross-slice attention is a compact and effective way to make 2.5D glioma segmentation more consistent through the volume. In the context of SDG 3, the demonstrated contribution is a lightweight model with better volumetric consistency; the reduction of clinicians' workload and diagnostic delay remains an expected benefit that requires clinical evaluation.

## 7.2 Future work

Several tasks remain for the next phase. The most immediate is to quantify robustness by retraining both models with several random seeds per fold, and to submit predictions to the official BraTS validation server so that results become directly comparable with published methods. A qualitative evaluation, with visual overlays of predictions and review by a radiologist, is needed to confirm that the measured gains in boundary accuracy are clinically meaningful. A 3D U-Net or nnU-Net reference trained under the same splits would show how much of the gap between 2.5D and 3D models the cross-slice module closes, and at what computational cost. The proposed data-size study, training both models on 16, 64, 150 and 265 patients with the corrected protocol, would test whether cross-slice context helps more when labelled data are scarce, as the preliminary mid-review study suggested; this is relevant for hospitals with few annotated scans. On the modelling side, the descriptor-based fusion of the original paper should be implemented as a second variant, and attention could be applied at the skip-connection levels as well as the bottleneck. Finally, evaluation on external and more recent datasets (such as later BraTS editions), robustness to missing modalities, and calibrated uncertainty estimates would be prerequisites for any clinical use.

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# Appendix A

# Source Code and Reproducibility

The complete source code is available in the public repository `https://github.com/sunilswain7/Glioma-Paper-Implementation` (folder `glioma_seg`). The mid-review prototype is preserved under the tag `v1-laptop`; the final system corresponds to commit `c0ac8eb` and later. The repository contains no patient data, credentials or trained weights; the trained checkpoints (about 0.9 GB) and the full results bundle are archived separately by the team.

The package `cst2` contains the final system: `preprocess.py` (NIfTI archive to per-patient arrays), `data.py` (patient-level folds and 2.5D stacks), `augment.py` (GPU augmentation), `model.py` (Cross-Slice U-Net), `losses.py`, `evaluate.py` (volumetric Dice, HD95, false-positive slices, ET post-processing), `train.py` and `tune.py` (Optuna search). The folder `astra` contains the cluster tooling: environment setup, experiment plans, the parallel job runner, the wrapper generator, the results collector and the bundling script. The folder `cst` holds the mid-review prototype.

## A.1 Setup and execution

On the Astra cluster the environment is created once from a Jupyter terminal; on other machines the same commands apply with a PyTorch build matching the local CUDA version.

```
git clone https://github.com/sunilswain7/Glioma-Paper-Implementation.git code
bash code/glioma_seg/astra/setup_astra.sh   # venv, torch 2.5.1+cu118,
                                            # numpy, scipy, nibabel, optuna
# dataset: Kaggle "awsaf49/brats20-dataset-training-validation" (~4.2 GB zip)
cd code/glioma_seg
python -m cst2.preprocess --zip ../../data/brats.zip \
       --out ../../data/brats_npy --workers 16
```

A single training run (here fold 0 of the proposed model) and the corresponding baseline are started as follows; each run writes `summary.json`, `history.json`, `test_per_patient.json`, `split.json` and `best.pth` to its run directory:

```
python -m cst2.train --data_dir DATA --out_dir RUNS --exp safe_k5_f0 \
       --n_slices 5 --fold 0 --epochs 30 --iters_per_epoch 400 --batch_size 32
python -m cst2.train --data_dir DATA --out_dir RUNS --exp safe_k1_f0 \
       --n_slices 1 --fold 0 --epochs 30 --iters_per_epoch 400 --batch_size 32
python -m cst2.tune  --data_dir DATA --out_dir RUNS --worker 0 --timeout_h 1
python astra/collect.py RUNS        # results table + paired Wilcoxon tests
```

On Astra, the plans in `astra/plans.py` group these runs into jobs. Running `astra/make_jobs.py` writes one scheduler wrapper per plan into `~/Project1/run_on_astra`, and each wrapper is submitted through *Jobs* → *Shell Script* with the GPU option enabled, queue `gpuq` and one GPU card. The main experiment of Chapter 6 corresponds to the plans `safety` (fold 0) and `safety_A`, `safety_B` and `safety_C` (folds 1–4).

## A.2 Supporting material

The configuration of every reported run is stored in its `summary.json`, and the patient identifiers of each split in `split.json`. Table A.1 lists the run directories behind the results in Chapter 6. The figures in this report are generated from the results bundle by `report/make_figures.py`.

Table A.1: Run directories behind the reported results

| **Runs** | **Content** |
|---|---|
| `safe_k{1,5}_f{0..4}` | Main five-fold comparison (Tables 6.1–6.4) |
| `tune_w*`, `optuna_*` | Optuna search (Table 6.5) |
| `long_k1_f*`, `long_kx_f*` | Supplementary long-training experiment (Table 6.6) |
| `final_k1_f{1,2}` | Short tuned-setting runs (terminated early to free GPUs; not used in conclusions) |